



AZURE LANDING ZONE

DOCUMENT CONTROL & METADATA

Document Title	Azure Landing Zone — Complete Enterprise SOP & Interview Master Guide
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Document Version	v2.0 (Official Production & Interview Edition)
Classification	Standard Operating Procedure (SOP) / Architecture Baseline
Target Audience	Cloud Architects, DevSecOps Engineers, Infrastructure Leads & Cloud Interview Candidates

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SECTION	TOPIC & CORE FOCUS	KEY ELEMENTS / PILLARS
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☀️ 1. Azure Landing Zone — Foundation

📖 1.1 Official Definition

An **Azure Landing Zone** is a proven and scalable Azure architecture that provides a governed, secure, and standardized foundation for deploying and operating workloads at scale.

Microsoft describes an Azure Landing Zone as a multi-subscription architecture designed to govern, secure, and scale Azure environments.

A Landing Zone consists of:

- 🛡️ **Platform Landing Zone:** Provides centralized capabilities such as governance, security, identity, networking, and management.
- 🚀 **Application / Workload Landing Zones:** Where workload teams deploy their applications and workloads within the guardrails defined by the platform.

The **Platform Landing Zone** provides centralized capabilities such as governance, security, identity, networking, and management.

The **Application Landing Zone** is where workload teams deploy their applications and workloads within the guardrails defined by the platform.

🎯 1.2 Interview Definition

"Azure Landing Zone is a standardized and governed Azure foundation that is prepared before application workloads are deployed. It provides common capabilities such as identity, networking, security, governance, monitoring, management, and cost controls so that workloads can be deployed consistently and securely at scale."

💡 1.3 Easy English

Imagine an organization wants to deploy **100 applications** in Azure.

If every application team independently creates:

- 🌐 VNet
- 🛡️ NSG
- 🔑 RBAC
- 📊 Logging
- 🚪 Security
- 📜 Policies
- 🗝️ Key Vault
- 📈 Monitoring
- 💾 Backup

then every team may implement them differently.

Landing Zone solves this problem.

We create the common foundation first.

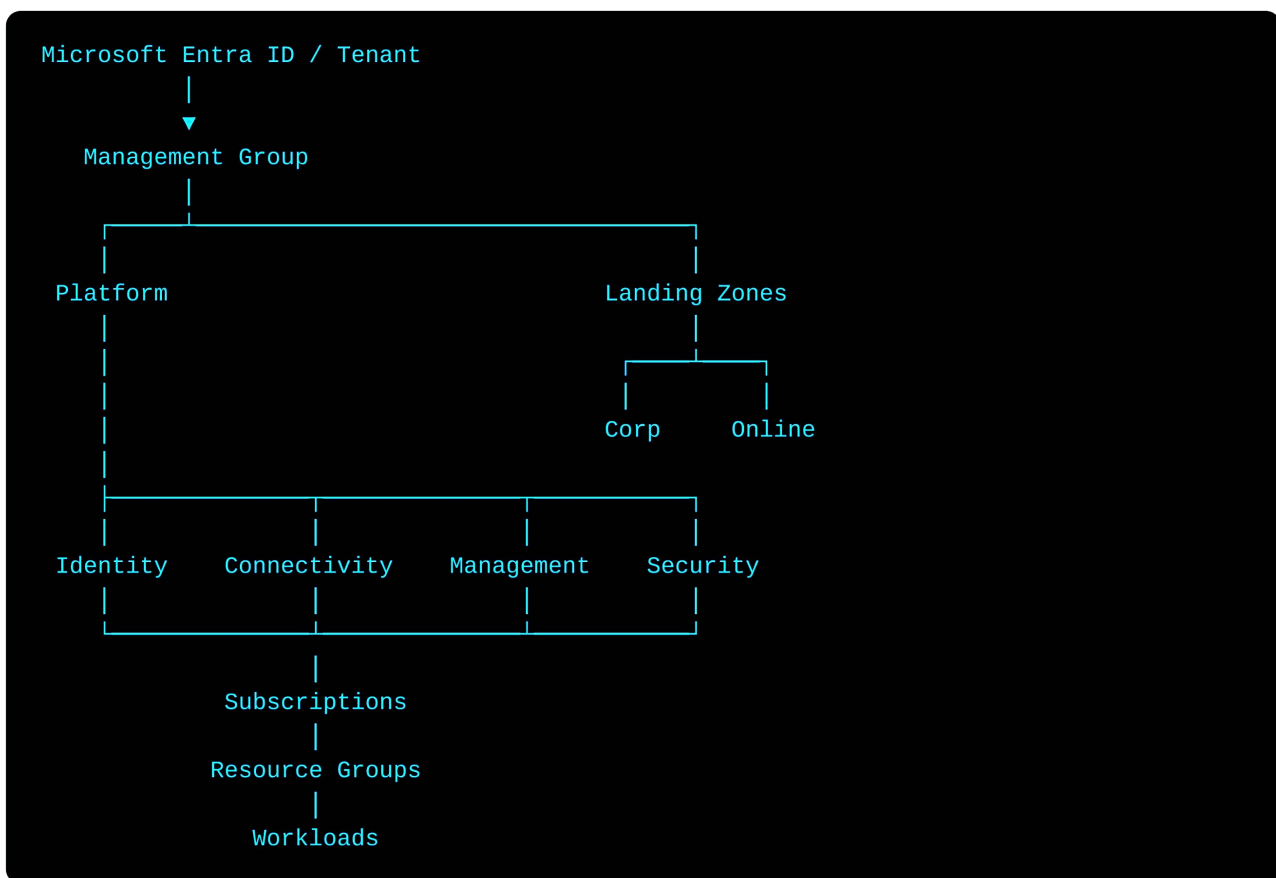
Then application teams deploy workloads on top of that foundation.

🎯 Simple Formula

```
Landing Zone
|
├─ Identity
├─ Hierarchy
```

- Governance
- Security
- Networking
- Compute
- Database
- Backup & DR
- Monitoring
- Cost Optimization

2. Landing Zone — High-Level Architecture



Microsoft's reference architecture uses Management Groups as the governance hierarchy and separates platform capabilities from application landing zones.

3. Nine Core Pillars

For our practical learning model, we will study Landing Zone through these pillars:

1.  Hierarchy
2.  Identity & Access Management
3.  Governance

4.  Security
5.  Networking & Connectivity
6.  Compute
7.  Database & Data
8.  Backup & Disaster Recovery
9.  Monitoring & Operations
10.  Cost Optimization

 **Note:** While traditionally categorized as "nine core pillars", including Cost Optimization makes 10 comprehensive categories. All 10 categories are retained in this SOP because cost management is a vital architectural design concern in production Landing Zones.

PILLAR 1 — HIERARCHY

1.1 Purpose of Hierarchy

The purpose of hierarchy is to logically organize Azure resources and apply governance from top to bottom through inheritance.

Basic hierarchy structure:

```
Tenant
|
▼
Management Group
|
▼
Subscription
|
▼
Resource Group
|
▼
Resource
```

1.2 Microsoft Entra Tenant

The Tenant represents the organizational identity boundary.

Example:

```
Company
|
Microsoft Entra Tenant
```

```
|
├── Users
├── Groups
├── Applications
└── Service Principals
```

1.3 Management Group

A Management Group logically groups multiple Azure subscriptions together.

Example:

```
Tenant Root Group
|
├── Platform
|   |
|   ├── Identity
|   ├── Connectivity
|   └── Management
|
├── Landing Zones
|   |
|   ├── Corp
|   └── Online
|
├── Sandbox
|
└── Decommissioned
```

The main purpose of Management Groups is to manage governance above the subscription level.

Policies and RBAC assignments applied at higher scopes automatically inherit down to all child resources.

In Microsoft reference architecture, dedicated Management Group structures such as Platform, Security, Management, Connectivity, and Identity are utilized.

1.4 Subscription

A Subscription serves as a critical boundary for billing, quotas, and access control.

Example:

```
Landing Zones
|
├── Production Subscription
├── Development Subscription
└── Testing Subscription
```

Keeping production environments in separate subscriptions provides:

- 🔒 Access isolation (Blast radius reduction)
- 💰 Granular cost tracking

- 📄 Independent governance
- 📊 Dedicated quota management
- 🛡️ Strict security boundaries

1.5 Resource Group

A Resource Group logically groups related Azure resources for lifecycle management.

Example:

```
rg-prod-payment
├── VM
├── NIC
├── NSG
├── Public IP
└── Key Vault
```

Important

A Resource Group is a **management and lifecycle boundary**, not a **network or security boundary**.

🔒 PILLAR 2 — IDENTITY & ACCESS MANAGEMENT

2.1 IAM (Identity & Access Management)

The core purpose of IAM is:

"Who can access what, and what specific actions can they perform?"

Main components:

```
Identity
├── User
├── Group
├── Service Principal
└── Managed Identity

├── RBAC
├── PIM
└── Access Reviews
```

2.2 User

Represents a human identity.

Example:

```
akmal@company.com
```

A user accesses Azure resources strictly based on assigned permissions.

2.3 Group

Simplifies permission management by grouping multiple users together.

Example:

```
DevOps-Team
├── User1
├── User2
└── User3

DevOps-Team
└── Contributor
```

Instead of assigning the Contributor role individually to 50 users, the role is assigned directly to the group.

2.4 RBAC (Role-Based Access Control)

Azure RBAC controls who can perform which actions on specific Azure resources.

Example:

```
User
├── Contributor
└── Resource Group
```

Common built-in roles:

- 👑 Owner: Full access to all resources including access delegation.
- 🛠 Contributor: Can create and manage all resources, but cannot grant access to others.
- 👁 Reader: Can view existing resources only.
- 👤 User Access Administrator: Manages user access to Azure resources.
- 📦 Storage Blob Data Contributor: Grants read/write/delete access to Azure Storage blob containers.
- 🔑 Key Vault Secrets User: Read secret contents from Key Vault.

RBAC Scope Levels

```
Management Group
|
Subscription
|
Resource Group
|
Resource
```

Best practice is to always assign RBAC at the minimum required scope following the principle of least privilege.

2.5 PIM (Privileged Identity Management)

Privileged Identity Management makes privileged administrative access Just-In-Time (JIT) and strictly controlled.

Instead of permanent admin access:

```
Admin
|
Permanent Owner
```

Use controlled JIT activation:

```
Admin
|
Eligible Status
|
Request Activation
|
Approval / MFA
|
Temporary Privileged Access (e.g., 4 Hours)
|
Access Expired
```

PIM is essential in production environments to eliminate permanent standing administrative privileges.

2.6 Managed Identity

Managed Identity allows Azure resources to authenticate securely with other Azure services without storing passwords, API keys, or secrets in application code or configuration.

Types:

```
Managed Identity
|
├── System Assigned (Tied to resource lifecycle; deleted automatically when
```

```
resource is deleted)
|
└─ User Assigned (Independent lifecycle; can be shared across multiple
resources)
```

Example:

```
VM
|
Managed Identity
|
RBAC
|
Key Vault
```

The Virtual Machine accesses Key Vault secrets directly and securely without any hardcoded credentials.

PILLAR 3 — GOVERNANCE

The purpose of governance is to make the Azure environment controlled, compliant, and standardized across the organization.

Main components:

```
Governance
|
├─ Azure Policy
|   └─ Built-in Policy
|       └─ Custom Policy
├─ Initiative
├─ Policy Assignment
├─ Compliance
├─ Tags
└─ Resource Naming
```

3.1 Azure Policy

Azure Policy continuously evaluates and enforces organizational rules across all resources.

Example:

```
Policy:
Only Allowed Locations
```

```
Allowed:
East US
West Europe
```

```
Not Allowed:
Any other random region
```

Policy Effects:

-  **Audit:** Generates a non-compliance warning without blocking resource creation.
-  **Deny:** Blocks resource deployment if it violates policy rules.
-  **Modify:** Automatically adds or updates required properties/tags during creation.
-  **Append:** Adds required fields to the resource payload.
-  **DeployIfNotExists:** Automatically deploys missing resources (e.g., auto-enabling diagnostic settings).
-  **AuditIfNotExists:** Audits if related child resources are missing.

3.2 Built-in Policy

Standard policies provided directly by Microsoft out-of-the-box.

Examples:

- Allowed locations for resource deployment
- Require specific tags on resources
- Require secure transfer (HTTPS) for storage accounts
- Allowed virtual machine SKUs

Built-in policies are ideal when standard organizational compliance rules already exist.

3.3 Custom Policy

Created for custom organization-specific business and security requirements.

Example:

```
Every resource deployed must contain an 'Environment' tag.
```

3.4 Initiative (Policy Set)

An Initiative is a collection of multiple policies grouped together for a single governance baseline.

Example:

```
Initiative: Production Security Baseline
|
└─ Require HTTPS
```

- └─ Require Mandatory Tags
- └─ Allowed Deployment Locations
- └─ Secure Storage Encryption
- └─ Disk Encryption for VMs

Simple Distinction:

Policy = One single rule
Initiative = A collection of multiple rules

3.5 Policy Assignment

Defining a policy is not enough; it must be assigned to a specific management scope.

```
Policy Definition
  |
  ▼
Assignment
  |
  ▼
Management Group / Subscription / Resource Group
```

Assigning Azure Policy at the Management Group level ensures seamless governance inheritance across all child subscriptions and resource groups.

PILLAR 4 — SECURITY

The purpose of security is to protect application workloads and the Azure foundation from cyber threats using defense-in-depth.

Key Security Controls:

```
Security
├─
├─ WAF (Web Application Firewall)
├─ Key Vault
├─ NSG (Network Security Group)
├─ ASG (Application Security Group)
├─ Private Endpoint
├─ Service Endpoint
├─ Defender for Cloud
├─ Azure Bastion
├─ Managed Identity
├─ DDoS Protection
└─ Azure Firewall
```

4.1 WAF (Web Application Firewall)

Protects web applications against common Layer 7 web vulnerabilities and attacks.

Traffic Flow:

```
Internet
|
Application Gateway
|
WAF (Inspection Layer)
|
Backend Application
```

Typical Protection:

- SQL Injection (SQLi)
- Cross-Site Scripting (XSS)
- OWASP Top 10 vulnerabilities and malicious HTTP requests

4.2 Key Vault

Securely stores and manages:

- 🗝️ Secrets (Passphrases, connection strings)
- 🔑 Keys (Encryption keys)
- 📜 Certificates (TLS/SSL certificates)

Example Architecture:

```
Application
|
Managed Identity (Passwordless)
|
RBAC Authentication
|
Key Vault
|
Secure Secret Retrieved
```

Golden Goal:

Never hardcode credentials or secrets in Terraform files, application source code, or CI/CD pipeline YAML.

4.3 NSG (Network Security Group)

Controls inbound and outbound network traffic filtering at Layer 4 (IP and Port level).

Example:

```
Internet
|
NSG (Rule Filtering)
|
Virtual Machine
```

Rules:

- Allow TCP 443 (HTTPS)
- Allow TCP 22 (SSH from authorized management subnet)
- Deny all unwanted inbound traffic

4.4 ASG (Application Security Group)

Logically groups Virtual Machine network interfaces (NICs) based on application roles.

Example:

```
ASG-Web
ASG-App
ASG-DB
```

Security Rule Logic:

```
ASG-Web → Allow Port 8080 → ASG-App
ASG-App → Allow Port 5432 → ASG-DB
```

Simplifies rule management by using logical role names instead of maintaining individual IP addresses.

4.5 Private Endpoint

Assigns a private IP address inside your Virtual Network to an Azure PaaS service via Azure Private Link.

Example:

```
VM
|
VNet
|
Private Endpoint
|
Private IP (e.g., 10.0.2.5)
|
Azure Storage / Key Vault / SQL Database
```

Public exposure can be completely disabled, ensuring zero internet exposure for PaaS data services.

4.6 Service Endpoint

Extends your VNet private identity to supported Azure services over the secure Microsoft Azure backbone network.

Example:

```
VM
|
VNet
|
Service Endpoint
|
Azure Storage (Microsoft Backbone)
```

Important Interview Distinction:

Private Endpoint = Private IP inside VNet + Azure Private Link (PaaS public IP disabled).

Service Endpoint = VNet identity routing over Azure backbone (PaaS service still retains public IP).

4.7 Defender for Cloud

Microsoft Defender for Cloud delivers comprehensive Cloud Security Posture Management (CSPM) and Cloud Workload Protection (CWPP).

Core Capabilities:

- Continuous security posture and Secure Score visibility
- Actionable security recommendations
- Advanced threat protection against malicious attacks
- Automated regulatory compliance benchmarking
- Multi-workload runtime protection (VMs, Containers, DBs)

4.8 Azure Bastion

Provides fully managed, browser-based secure RDP/SSH connectivity to VMs without requiring public IP addresses on VMs.

```
Administrator
|
Azure Portal (HTTPS 443)
|
Azure Bastion Subnet
|
Private IP (Port 22/3389)
|
Target Virtual Machine
```

4.9 Managed Identity

From a security perspective, Managed Identity eliminates the need to store credentials in code or configuration files.

Example:

```
VM
|
System-Assigned Identity
|
RBAC Role Assignment
|
Storage Account
```

PILLAR 5 — NETWORKING & CONNECTIVITY

Networking is the backbone and core operational component of an Azure Landing Zone.

```
Networking
|
├── VNet
├── Subnet
├── NSG
├── ASG
├── Hub-Spoke
├── VNet Peering
├── Azure Firewall
├── Application Gateway
├── Load Balancer
├── Private Endpoint
├── Service Endpoint
├── VPN Gateway
├── ExpressRoute
├── Azure Bastion
└── DNS
```

5.1 Hub-and-Spoke Topology

Centralizes shared networking and security services in the **Hub VNet**, while isolating application workloads in dedicated **Spoke VNets**.



Spoke-1 (Payment) Spoke-2 (Portal) Spoke-3 (Analytics)

The Hub VNet typically contains:

- Azure Firewall (Central traffic inspection)
- VPN Gateway (Site-to-Site tunnels)
- ExpressRoute Gateway (Private enterprise link)
- Azure Bastion Host
- Private DNS Resolvers
- Shared network management services

5.2 VNet Peering

Connects virtual networks privately with high bandwidth and ultra-low latency over the Microsoft backbone.

VNet-A $\longleftrightarrow$ Non-Transitive Peering $\longrightarrow$ VNet-B

5.3 Azure Firewall

A cloud-native, fully stateful, centralized network security service with built-in high availability.

Example Flow:

```
Spoke Workload
|
Hub VNet
|
Azure Firewall (Deep Inspection)
|
Internet / On-Premises
```

Firewall Capabilities:

- Network filtering rules (L3/L4)
- Application filtering rules (FQDN / URL inspection)
- Source and Destination NAT rules
- Centralized logging and traffic inspection

5.4 Application Gateway

Layer 7 application load balancer that manages web application traffic intelligently.

Key Features:

- HTTP/HTTPS intelligent routing
- SSL/TLS termination and offloading
- Integrated Web Application Firewall (WAF)

- URL path-based routing
- Multi-site / Host-based routing

5.5 Azure Load Balancer

High-throughput, ultra low-latency Layer 4 (TCP/UDP) load balancer.

```
Client Request
|
Azure Load Balancer (Layer 4)
|
├── VM 1 (Healthy Backend)
|
└── VM 2 (Healthy Backend)
```

5.6 VPN Gateway

Provides encrypted cross-premises network connectivity between Azure and on-premises environments over the public internet.

```
On-Premises Data Center
|
IPSec VPN Tunnel (Encrypted)
|
Azure VPN Gateway
|
Virtual Network
```

5.7 ExpressRoute

Provides dedicated, private, ultra low-latency connectivity between on-premises infrastructure and Microsoft Cloud.

Used when organizations require enterprise-grade private connectivity, high reliability, and predictable networking performance without traversing the public internet.

PILLAR 6 — COMPUTE

Compute represents the hosting environment where application workloads actually execute.

```
Compute
|
├── Virtual Machine (VM)
├── VM Scale Sets (VMSS)
├── App Service
└── AKS (Azure Kubernetes Service)
```

```
└─ Container Apps
└─ Azure Functions
```

6.1 Virtual Machine (VM)

Infrastructure as a Service (IaaS) compute offering complete administrative control.

Customer Responsibilities:

- Operating System installation & configuration
- OS patching and security maintenance
- Application runtime deployment
- Subnet and network configuration

6.2 VM Scale Sets (VMSS)

Deploys and manages a group of identical, load-balanced virtual machines with automated scaling capabilities.

```
VMSS Controller
├─
├─ VM Instance 1
├─ VM Instance 2
└─ VM Instance 3 (Auto-scaled on CPU demand)
```

6.3 App Service

Fully managed Platform as a Service (PaaS) for building, hosting, and scaling web applications and REST APIs.

Eliminates underlying OS management and infrastructure maintenance overhead.

6.4 AKS (Azure Kubernetes Service)

Enterprise managed Kubernetes orchestration service.

Ideal for complex containerized microservices requiring dynamic orchestration, autoscaling, and self-healing capabilities.

PILLAR 7 — DATABASE & DATA

Azure provides comprehensive managed data and storage solutions tailored for diverse workload needs:

```
Database & Data
├─
└─ Azure SQL Database
```

```
└─ SQL Server on VM
└─ Azure Database for PostgreSQL
└─ Azure Cosmos DB
└─ Azure Database for MySQL
└─ Azure Storage Accounts
```

7.1 Azure SQL Database

Fully managed, intelligent relational database engine with automated patching, backups, and built-in high availability.

Secure Application Pattern:

```
App Service / VM
|
Private Endpoint (Private IP)
|
Azure SQL Database (Public Access Disabled)
```

7.2 SQL Server on Azure VM

Traditional SQL Server running inside an Azure IaaS VM.

Provides complete OS and DB administrative control with higher management overhead compared to managed PaaS SQL.

7.3 Azure Database for PostgreSQL

Fully managed PostgreSQL database service with automated maintenance and enterprise security.

Widely used for open-source applications requiring native PostgreSQL compatibility.

7.4 Azure Cosmos DB

Globally distributed, multi-model NoSQL database service designed for modern cloud apps.

Key Strengths:

- Global distribution and turnkey multi-region replication
- Guaranteed single-digit millisecond latency
- Flexible schema-less data modeling
- Highly scalable for distributed modern applications

7.5 Azure Storage Account

Provides massively scalable storage for:

- Blob: Unstructured object storage for files, images, and backups
- Files: Fully managed SMB/NFS cloud file shares
- Queues: High-volume message queuing between decoupled services

- Tables: NoSQL key-value data storage

Security Architecture:

```
Storage Account
|
Private Endpoint
|
Virtual Network Subnet
|
Application Workload
```



PILLAR 8 — BACKUP & DISASTER RECOVERY

In enterprise Landing Zones, Backup and Disaster Recovery (DR) are mandatory architectural requirements to ensure business continuity.

```
Backup & DR
|
├─ Azure Backup
├─ Recovery Services Vault
├─ Azure Site Recovery (ASR)
├─ Availability Zones
├─ Geo-Redundancy
├─ RPO (Recovery Point Objective)
└─ RTO (Recovery Time Objective)
```

8.1 Azure Backup

Simple, secure, and cost-effective centralized backup solution for data and resource recovery.

Example Flow:

```
Virtual Machine
|
Azure Backup Engine
|
Recovery Services Vault (Encrypted Retention)
```

8.2 Recovery Services Vault

A centralized storage entity in Azure used to manage and retain backup data and disaster recovery replication policies.

8.3 Azure Site Recovery (ASR)

Orchestrates business continuity and disaster recovery by automating cross-region replication, failover, and failback scenarios.

```
Primary Azure Region (Production)
|
| Continuous Asynchronous Replication
|
▼
Secondary Azure Region (DR Target)
```

8.4 RPO (Recovery Point Objective)

Recovery Point Objective defines:

The maximum acceptable amount of data loss measured in time.

Example:

```
RPO = 15 Minutes
```

This means the organization can tolerate a maximum data loss of up to 15 minutes of transactional data during an outage.

8.5 RTO (Recovery Time Objective)

Recovery Time Objective defines:

The maximum acceptable duration of time required to restore service availability after a disruption.

Example:

```
RTO = 1 Hour
```

PILLAR 9 — MONITORING & OPERATIONS

The purpose of monitoring is to gain full visibility into the health, performance, security, and operational availability of all cloud workloads.

```
Monitoring
|
├─ Azure Monitor
├─ Log Analytics
```

```
├─ Application Insights
├─ Alerts
├─ Action Groups
├─ Metrics
├─ Logs
├─ Microsoft Sentinel
└─ Network Watcher
```

9.1 Azure Monitor

Comprehensive monitoring service for collecting, analyzing, and acting on telemetry from cloud and on-premises environments.

Collects:

- Real-time performance metrics
- Detailed operational logs
- Automated alert notifications
- Diagnostic settings data

9.2 Log Analytics Workspace

Centralized repository for aggregating, indexing, and querying log data across all Azure subscriptions.

Example:

```
VM / PaaS Service
|
Diagnostic Settings Export
|
Log Analytics Workspace
|
KQL Queries / Dashboards / Alert Rules
```

9.3 Application Insights

Application Performance Management (APM) service for live web applications and microservices.

Monitors:

- Request rates, response times, and failure rates
- External dependency call rates and performance
- Exceptions and trace logs
- Real-time user telemetry and synthetic availability tests

9.4 Alerts & Action Groups

Automatically triggers proactive notifications or remediation scripts when specific threshold conditions are met.

Example:

```
CPU Usage > 80% for 5 mins
|
Alert Triggered
|
Action Group
|
Email / SMS Notification / PagerDuty / Webhook Automation
```

9.5 Microsoft Sentinel

Cloud-native Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) solution.

Enterprise Flow:

```
Azure Resources & Identity Events
|
▼
Diagnostic Logs
|
▼
Log Analytics Workspace
|
▼
Microsoft Sentinel (AI Threat Detection)
|
▼
Security Incident Investigation & Automated Playbooks
```

PILLAR 10 — COST OPTIMIZATION

An enterprise Landing Zone must not only be secure and compliant, but also highly cost-efficient and financially transparent.

```
Cost Optimization
|
├─ Cost Management
├─ Budgets
├─ Tags
├─ Reservations
├─ Savings Plans
├─ Right Sizing
├─ Auto Shutdown
├─ Environment Separation
└─ Cost Allocation
```

10.1 Cost Management

Provides unified visibility to analyze, monitor, and optimize cloud spending across subscriptions and departments.

10.2 Budgets & Thresholds

Enables teams to define monthly financial thresholds and configure proactive alerts before budget overruns occur.

Example:

```
Monthly Budget = $5,000
```

```
80% Consumption → Email Alert to Team Lead
```

```
90% Consumption → Warning to Department Head
```

```
100% Consumption → Executive Critical Alert
```

10.3 Governance Tagging for Cost Attribution

Tags provide essential metadata for accurate cost center allocation and governance tracking.

Standard Tagging Schema:

```
Environment = Production
Application = PaymentGateway
Owner       = DevOpsTeam
CostCenter  = CC101
Project     = BankingCore
```

Microsoft Well-Architected Framework recommends enforcing mandatory tagging policies to support robust FinOps practices and chargeback models.



COMPLETE LANDING ZONE STRUCTURE

AZURE LANDING ZONE

- |
- |— 1. HIERARCHY
 - | |— Tenant Root Group
 - | |— Management Groups (Platform & Landing Zones)
 - | |— Subscriptions (Corp, Online, Shared)
 - | |— Resource Groups
- |
- |— 2. IDENTITY & ACCESS MANAGEMENT
 - | |— Microsoft Entra ID
 - | |— Users & Security Groups
 - | |— Service Principals & App Registrations
 - | |— Managed Identities (System & User Assigned)

- | | | | | Role-Based Access Control (RBAC)
- | | | | | Privileged Identity Management (PIM)
- | | | | | Access Reviews & Governance

| | | | | 3. GOVERNANCE

- | | | | | | | Azure Policy Definitions
 - | | | | | | | | Built-in Policies
 - | | | | | | | | Custom Policies
- | | | | | | | Policy Initiatives (Policy Sets)
- | | | | | | | Scoped Policy Assignments
- | | | | | | | Compliance Dashboard
- | | | | | | | Mandatory Tagging Standards

| | | | | 4. SECURITY

- | | | | | | | Web Application Firewall (WAF)
- | | | | | | | Azure Key Vault (HSM-backed)
- | | | | | | | Network Security Groups (NSG)
- | | | | | | | Application Security Groups (ASG)
- | | | | | | | Private Endpoints & Private Link
- | | | | | | | Service Endpoints
- | | | | | | | Microsoft Defender for Cloud (CSPM & CWPP)
- | | | | | | | Azure Bastion Host
- | | | | | | | Managed Identity (Zero Trust)
- | | | | | | | Azure Firewall
- | | | | | | | DDoS Network Protection

| | | | | 5. NETWORKING & CONNECTIVITY

- | | | | | | | Virtual Networks (VNet) & Subnets
- | | | | | | | Hub-and-Spoke Architecture
- | | | | | | | Virtual Network Peering
- | | | | | | | Azure Firewall (Central Egress Inspection)
- | | | | | | | Application Gateway (Layer 7 WAF)
- | | | | | | | Azure Load Balancer (Layer 4)
- | | | | | | | VPN Gateway (IPSec Hybrid)
- | | | | | | | Azure ExpressRoute (Dedicated Fiber)
- | | | | | | | Private Endpoints
- | | | | | | | Service Endpoints
- | | | | | | | Azure Bastion Subnet
- | | | | | | | Private DNS Zones & Resolvers

| | | | | 6. COMPUTE

- | | | | | | | Virtual Machines (IaaS)
- | | | | | | | Virtual Machine Scale Sets (VMSS)
- | | | | | | | Azure App Service (PaaS)
- | | | | | | | Azure Kubernetes Service (AKS)
- | | | | | | | Azure Container Apps
- | | | | | | | Azure Functions (Serverless)

| | | | | 7. DATABASE & DATA

- | | | | | | | Azure SQL Database
- | | | | | | | SQL Server on Azure VMs
- | | | | | | | Azure Database for PostgreSQL
- | | | | | | | Azure Cosmos DB (Multi-Region NoSQL)
- | | | | | | | Azure Database for MySQL

```

└─ Azure Storage Accounts (Blob, File, Data Lake)
└─ 8. BACKUP & DISASTER RECOVERY
    └─ Azure Backup Engine
    └─ Recovery Services Vault
    └─ Azure Site Recovery (ASR Failover)
    └─ Availability Zones (Multi-Zone HA)
    └─ Geo-Redundant Storage (GRS)
    └─ Recovery Point Objective (RPO)
    └─ Recovery Time Objective (RTO)
└─ 9. MONITORING & OPERATIONS
    └─ Azure Monitor Platform
    └─ Central Log Analytics Workspace
    └─ Application Insights (APM)
    └─ Metric & Log Alert Rules
    └─ Action Groups (Automated ITSM)
    └─ Metrics Explorer
    └─ Azure Activity & Resource Logs
    └─ Microsoft Sentinel (Cloud SIEM/SOAR)
    └─ Network Watcher & Traffic Analytics
└─ 10. COST OPTIMIZATION & FINOPS
    └─ Microsoft Cost Management
    └─ Budgets & Threshold Alerts
    └─ Mandatory FinOps Cost Tags
    └─ Azure Reservations (1 & 3 Year)
    └─ Azure Savings Plans
    └─ Compute Right-Sizing
    └─ Departmental Cost Allocation

```

REAL ENTERPRISE EXAMPLE

Suppose an enterprise organization has:

- 100 Diverse Business Applications
- 10 Autonomous Development Teams
- 5 Dedicated Production Operations Teams
- Multiple Multi-region Azure Subscriptions
- Existing On-Premises Data Center Footprint

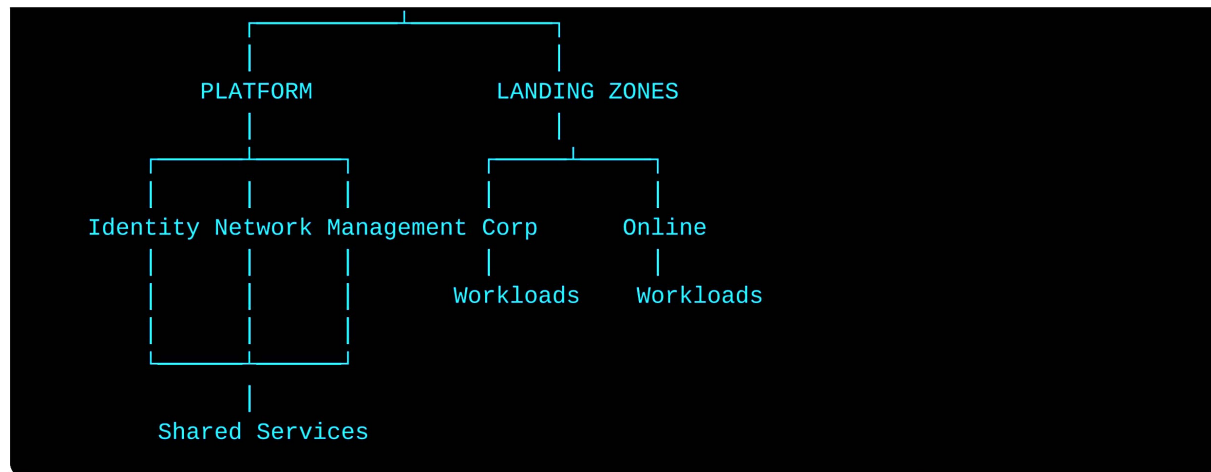
Directly deploying applications without a foundation creates massive operational risk and security vulnerabilities.

Instead, deploy the Landing Zone Foundation:

```

      AZURE TENANT
        |
    MANAGEMENT GROUP
        |

```



Central Platform Team Responsibilities:

- Identity and access management guardrails
- Centralized hub networking and perimeter security
- Enterprise security controls and Key Vault policies
- Global governance and Azure Policy enforcement
- Centralized monitoring, logging, and SIEM integration
- Shared infrastructure services

Workload Application Team Responsibilities:

- Application business logic and source code
- Application compute instances within provisioned subnets
- Workload-specific databases and storage containers
- Application-specific resources inside guardrails

Microsoft specifically recommends this clean operational separation between the Platform Landing Zone and Application Landing Zones.



ACTUAL STEP-BY-STEP LANDING ZONE FLOW

```
graph TD; Step1[STEP 1: Gather Enterprise & Regulatory Requirements] --> Step2[STEP 2: Design Management Group Hierarchy]; Step2 --> Step3[STEP 3: Establish Identity, RBAC & PIM Controls]; Step3 --> Step4[STEP 4: Deploy Hub-Spoke Network Foundation]; Step4 --> Step5[STEP 5: Configure Security Controls & Key Vaults];
```

The flowchart outlines the five steps for creating a landing zone:

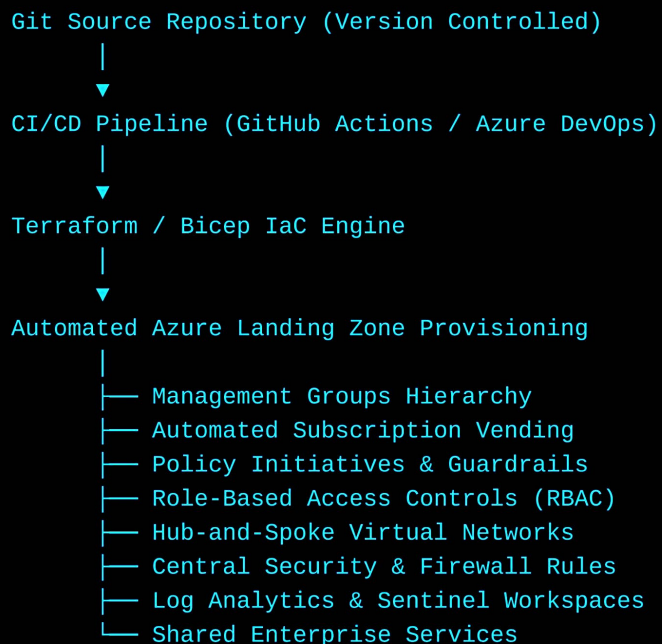
- STEP 1: Gather Enterprise & Regulatory Requirements
- STEP 2: Design Management Group Hierarchy
- STEP 3: Establish Identity, RBAC & PIM Controls
- STEP 4: Deploy Hub-Spoke Network Foundation
- STEP 5: Configure Security Controls & Key Vaults



⚙️ DEVOPS & TERRAFORM AUTOMATION PERSPECTIVE

In enterprise cloud environments, building Landing Zones manually through the Azure Portal is slow, error-prone, and non-scalable.

Enterprise Automation Pipeline:



Using Infrastructure as Code (IaC), the entire enterprise Landing Zone is codified, repeatable, testable, and version-controlled.

Microsoft officially documents Terraform and Bicep-based Subscription Vending modules for automated enterprise scale.

INTERVIEW MASTER ANSWER

Question:

What is an Azure Landing Zone?

Interview Master Answer (English):

"An Azure Landing Zone is a standardized, secure, and governed Azure foundation designed to deploy and operate workloads at enterprise scale. Before deploying applications, we establish core shared capabilities including identity and access management, management group hierarchies, hub-spoke networking, perimeter security, governance policies, centralized monitoring, backup and disaster recovery, and cost management controls. The Platform Landing Zone provides centralized shared services and governance guardrails, while Application Landing Zones allow workload teams to deploy their applications autonomously and securely within those guardrails."

Simplified Conceptual Explanation:

"In simple terms, an Azure Landing Zone is like preparing the complete infrastructure of a smart city—roads, electricity, water, security, and municipal laws—before constructing residential houses. The city provides the shared security and utilities (Platform), and individual families can build their homes safely on their assigned plots (Workloads) without building their own power plants or security forces from scratch."

INTERVIEW QUICK DIFFERENCE CHEATSHEET

CONCEPT	ONE-LINER ENTERPRISE MEANING
Tenant	Top-level organizational Identity boundary (Microsoft Entra ID).
Management Group	Subscription organization and policy/RBAC inheritance hierarchy.
Subscription	Billing, quotas, and access blast-radius boundary.
Resource Group	Logical lifecycle grouping of related Azure resources.
RBAC	Controls who can perform which specific actions on resources.

CONCEPT	ONE-LINER ENTERPRISE MEANING
PIM	Provides Just-In-Time (JIT) eligible temporary privileged access.
Policy	A single governance and compliance evaluation rule.
Initiative	A collection of grouped policies for a security baseline.
NSG	Layer 4 packet filtering by IP address and port numbers.
ASG	Logical grouping of VM NICs by application tier/role.
Private Endpoint	Assigns a private VNet IP to PaaS services via Azure Private Link.
Service Endpoint	Routes VNet traffic over Azure backbone; PaaS still has public IP.
WAF	Layer 7 web application firewall protecting against HTTP/S attacks.
Azure Bastion	Secure browser-based SSH/RDP to VMs without public IP exposure.
Managed Identity	Passwordless authentication between Azure resources without secrets in code.
Azure Firewall	Centralized, stateful network firewall for enterprise traffic inspection.
Azure Monitor	Unified telemetry and observability platform.
Log Analytics	Centralized repository for querying and analyzing log data.
Microsoft Sentinel	Cloud-native SIEM and SOAR intelligent security engine.
Azure Backup	Point-in-time data and VM snapshot protection and restore.
Site Recovery (ASR)	Automated cross-region replication and disaster recovery failover.
Cost Management	Cloud cost visibility, budgeting, and FinOps allocation.



MOST IMPORTANT INTERVIEW CONCEPT

Landing Zone ≠ Just a VNet

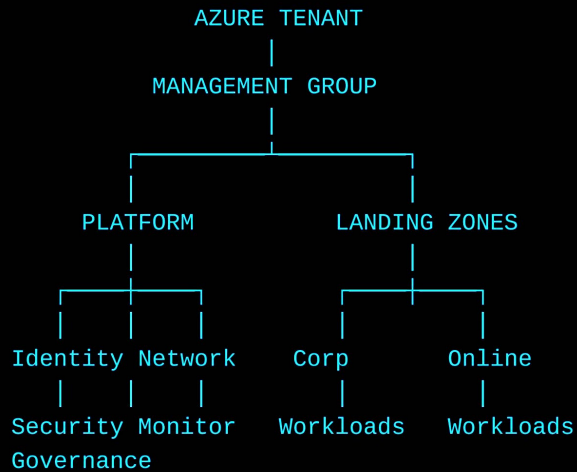
Landing Zone ≠ Just a Subscription

Landing Zone ≠ Just a Terraform Script

```
Landing Zone =
Governance Guardrails
+ Identity & Access Controls
+ Management Hierarchy
+ Scalable Networking
+ Enterprise Security
```

- + Operations Management
- + Workload Foundation
- + Observability
- + FinOps & Cost Controls

Core Enterprise Architecture Blueprint:



✨ Golden Rule to Remember:

"First build the foundation, then deploy the workload."

This is the core philosophy and essence of an Azure Landing Zone.